Cube and Cuboid

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We see a lot of things like book, pen, pencil, rubber etc. around us. These are 3D objects but we can see 2D shapes like triangle, quadrilateral, circle etc. in them. The surface, edges and vertices of these 3D objects are made up by combining these 2D shapes. We will now learn about surface area and volume of 3D objects such as cube and cuboid. To do this we will use the concept of 2D shapes and their area.

Representing 3D Shapes

We can draw 2D shapes (triangle, quadrilateral and circle etc.) on paper as per their measures.

Can we similarly depict three dimentional shapes on paper with their measurements?

Representing A Cube

While depicting 3D shapes we try to show all the faces. Those on the front as well on those of behind the faces in front. Cubes and Cuboids have 6 faces. They have 8 edges. How many vertices do they have?

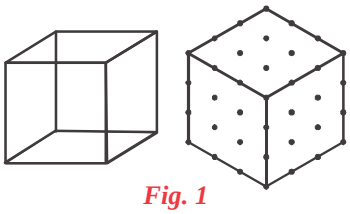


Fig. 1

Opening A Three Dimensional Shape to make a 2D Shape

1. Surface Area of A Cube:

Take a closed chalk box and open it as shown in the figure.

Now write down the number of faces, vertices and edges. Are all the faces congruent?

We find that all the faces of chalk box are of the same size, i.e. they are all congruent. Such a shape is called a cube.

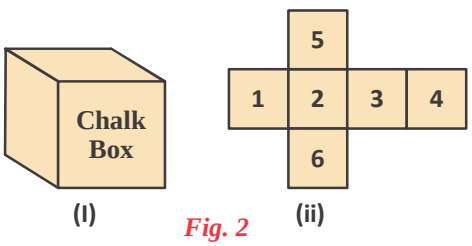


Fig. 2

If the length of the side of a square is 'a' then it's area would be a^2.

Can you find out the area of all the surfaces?

Hamid- I can add the area of 6 faces and find the total surface area.

Total surface area of the cube $= a^2 + a^2 + a^2 + a^2 + a^2 + a^2$

$= 6\ a^2$

Try This

1. Take a card board sheet and make a cube of side 8 *cm*.
2. The length of the edge of a cube is 4 *cm.* By how many times will the total surface area increase, if the length of the edge is increased to 8*cm.*



2. Cuboid

Take a box of toothpaste and open it as shown in the picture.



Fig. 3 (i)



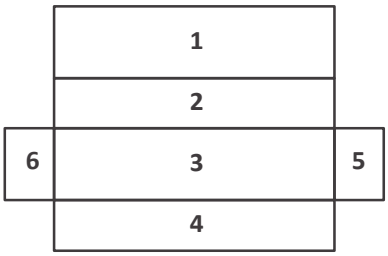


Fig. 3 (ii)

Now write down the number of faces, edges and vertices it has.

Are the lengths and breadths of faces 1 and 3, 2 and 4 and 5 and 6 the same?

We see that for a toothpaste box, the opposite faces have the same measure i.e. they are congruent. Such a shape is called a cuboid.

Surface Area of a Cuboid

Locate some cuboid shaped objects around you and draw their figures in your notebook. Name their vertices.

How many sides does each have (*Fig.*4)?

Which faces of the cuboid are equal to each other and rectangular?

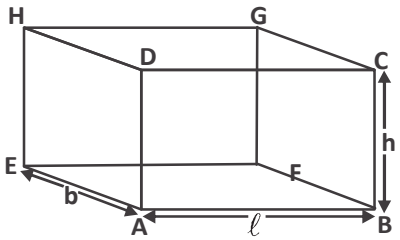


Fig. 4

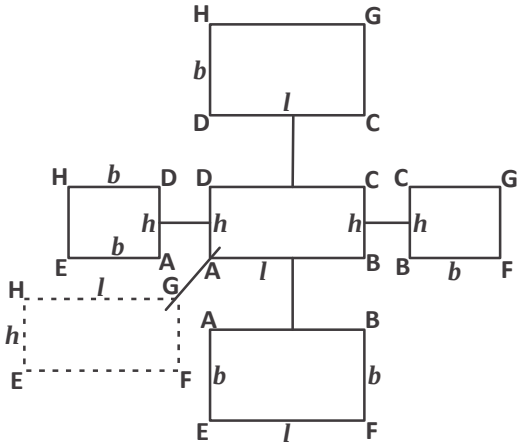


Fig. 5

How can you find the total surface area of a cuboid? Jaspal said that we can obtain the total surface area by adding area of all the rectangular surfaces.

Neha calculate the total surface area of a shoe-box.

She wrote l for length, b for breadth and h for height on the surfaces. Then she calculated the area of each surface seperately and added them. (*Fig.*5).

You also do the above task and see if you get the same.

So the total surface area of a cuboid = hl + lb + hl + lb + hb + hb

= 2 lb + 2 lh + 2 bh

= 2 (lb + lh + bh)

Try This



1. How will you find the total surface area of a cube using the formula for a cuboid?
2. What is the total surface area of a cuboid whose length is 6 *cm*, breadth is 3 *cm* and height is 2 *cm*.

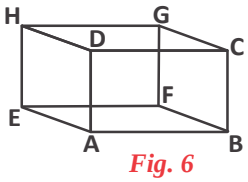


Fig. 6

Do you know which of the faces of the cuboid are the lateral faces?

The four side faces of the cuboid (excluding the top and bottom faces) are the lateral faces.

How will you find the surface area of the lateral faces of a cuboid?

Ajit- I can add the areas of faces ABCD, EFGH, BFGC and AEHD and that is the area of the lateral faces of the cuboid.

Monika- Yes, if we do this, the area of lateral faces of a cuboid = 2hl + 2 hb

= 2 h (l + b)

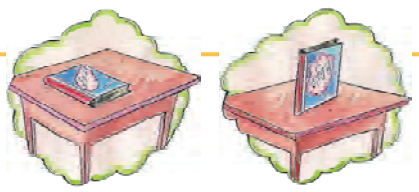
Can you now find the area of the lateral faces of a cube?

The area of lateral faces of a cube = 4 a^2 (why?)

Think and Discuss



Keep your mathematics book as shown in the pictures and find out its total surface area (Add the third position yourself).



Volume of Solid Objects



Fig. 7

Take a glass tumbler, fill it with water till the top. Now put 2 pieces of lemon in it. What do you see? Some water will flow out of it. This tells us that lemon takes up some of the space in the tumbler. In the same way all solid objects occupy space.

If a hollow object filled with a fluid like water or air then the fluid will take the shape of that object. In that case the volume of the fluid gives us the volume of that object or capacity of the object.

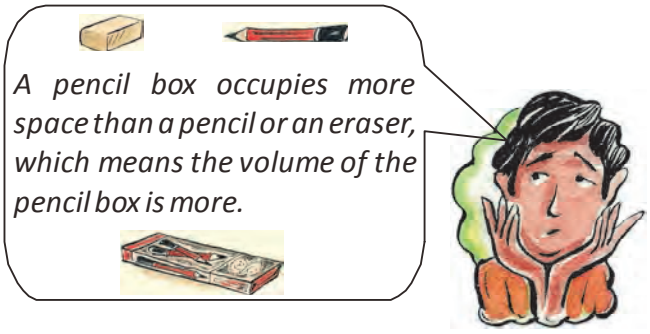
The milk pot (patila) can contain more milk then in the bowl (katori). Is the capacity of the patila more than that of the katori.



Volume of Cuboid

Is the volume of the room more than the volume of the almirah kept in the room? Or which has more volume, the pencil box or the pencils and eraser kept in it. Can you find the volume of some of these objects?

We use the unit square to measure the area of any surface, similarly we use the unit cube to find the volume of a solid because the cube is the simplest solid shape. To find the area we divide the surface into unit squares. Similiarly, to find the volume of a solid we will divide it into unit cubes.



1 cubic centimeter $= 1\ cm \times 1\ cm \times 1\ cm = 1\ cm^3$

1 cubic meter $= 1\ m \times 1\ m \times 1\ m = 1\ m^3$

Volume of a cuboid - surface area of the rectangular base × height

$= A \times h = l \times b \times h$ unit cubes

Here A is the area of the surface of the base of the cuboid and h is its height.

Volume of Cube

We know that cube is a special kind of cuboid whose length, breadth and height are equal i.e. $l = b = h = a$ (suppose)

Volume of cube = Side × Side × Side

$= a \times a \times a$

$= a^3$ cubic unit

Where a is the side of the cube.

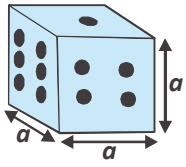


Fig. 8

To find the volume of a solid object using unit cubes is very convenient. The meaning of cubic unit of volume is number of unit length cubes included in it.

EXAMPLE-1. A cuboidal room has a length of 12 meter, breadth of 8 meter and height 4 meters. If the rate of white washing is Rs. 7 per sq m then how much money will be spent in painting its 4 walls and the ceiling?

SOLUTION : The length of the room $l = 12\ m$, breadth $b = 8\ m$ and height $h = 4\ m$

Since the room is a cuboid therefore,

the area of four walls of the room = perimeter of the base × height of the room

$= 2\ (l + b) \times h$

$= 2\ (12 + 8) \times 4 = 160$ sq m

Area of the ceiling $= l \times b$ $= 12 \times 8 = 96$ sq m

$\therefore$ Total area which needs to be while washed = 160 + 96 = 256 sq m

The rate of white washing is Rs. 7 per Sq. meter, therefore the cost of white washing the walls and the ceiling,

= Total area to be white washed × 7

= 256 × 7 = Rs. 1792

Try This



Measure the length, breadth and height of your classroom with your friends. Then find out:-

(i) The total surface area of the walls excluding the windows and doors.

(ii) If the room is to be white washed, then what is the total area that will need to be white washed.

(iii) What is the rate for white washing in your town/village? Using that what would be expenditure on getting the walls of the room white washed.

EXAMPLE-2. Anwar got a cubical water tank (with a lid) constructed on the terrace of his house. The length of outer edge 1.8 meter. He wants to put square tiles of side 30 *cm* on the entire surfase barring the base of the tank. If the cost of putting a dozen tiles is Rs 396, then what is the total expenditure that Anwar will incur?

SOLUTION : Since Anwar wants to get the five outer faces of the water tank tiled, therefore to find the number of tiles we need to find the surface area of these five surfaces.

Outer length of the cubical tank $a = 1.8\ m = 180\ cm$

$\therefore$ The area of the five faces of the tank $= 5\ a^2$

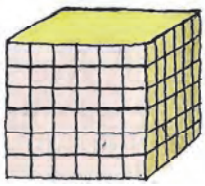
$= 5 \times 180 \times 180\ sq\ cm$ .....(1)

Area of each square tile $=$ side $\times$ side

$= 30 \times 30\ sq\ cm$ .....(2)

$$\therefore \text{ No. of tiles } = \frac{\text{Area of the five faces of the tank}}{\text{Area of each tile}}$$

$$= \frac{5 \times 180 \times 180}{30 \times 30} = 180$$



Cost of putting one dozen tiles = Rs. 396

$$\therefore \text{ Cost of putting one tile } = \frac{396}{12} = \text{Rs. } 33$$

So, the cost of putting 180 tiles = 180×33 = Rs. 5940

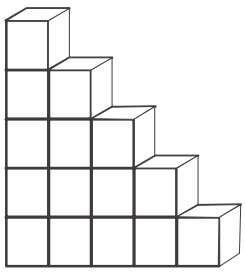
EXAMPLE-3. While playing with some cubical blocks, William made the structure shown in the picture. If the side of each cube is 4 *cm*, then find the volume of the structure made by Willam.

SOLUTION : Since the length of each side of cube, $a = 4\ cm$

$\therefore$ The volume of each cube $= a \times a \times a$

$= 4 \times 4 \times 4$

$= 64\ cubic\ cm$



Cubes used in the structure = 15

$\therefore$ The volume of the structure = $64 \times 15 = 960\ cubic\ cm$

Try This

Find the area of any rectangular sheet of your book. Now measure the height of the book and thus find its volume. In a shelf of an almirah in your school's library, how many such books can be placed? Do you read newspaper in your school every day? Now instead of the book, consider the almirah is filled with newspaper and do the same estimation and find out in how many days the almirah will be filled with newspapers?



Exercise - 15.1



1. The total surface area of a cube is 1350 m^2, find its volume.
2. The volume of a cuboid is 1200 cm^3. Its length is 15 *cm,* breadth 10 *cm* Find out its height?
3. What would the total surface area of a cubical box be if :-
   (i) each side is doubled?
   (ii) each side is tripled?
   (iii) each side is made *n* times?
4. The perimeter of the largest room in Priyanka's house is 250 *m.* The cost of white washing its four walls at the rate of Rs. 10 per sq m cost is Rs. 15000. What is the height of the room. (***Hint*** : Area of four walls = lateral surface area)
5. The length of the side of a cubical box is 10 *cm.* There is another cuboidal box with its length, breadth and height being 12.5 *cm*, 10 *cm* and 8 *cm* respectively.
   (i) Which box has more lateral surface area and by how much is it more?
   (ii) Which box has less total surface area and how much less?
6. The population of a village is 4000. Each person in the village requires 150 litre of water every day. There is a water tank in the village with length, breadth and height 20 *m*, 15 *m* and 6 *m* respectively. If this tank is filled to the brim with water then for how many days, the water requirement would be met?
7. A river 3 meter deep and 40 meter wide is flowing at the rate of 2 *km* per hour. Can you find out the volume of water going to the sea per minute?
8. The total surface area of a cuboid is 3328 m^2. If the length, breadth and height are in the ratio 4 : 3 : 2, then what will be its volume?
9. A cuboid is made by joining the edges of three identical cubes each having a volume of 125 cm^3. Find the total surface area of the cuboid thus formed?
10. A water body is of cuboidal shape of (Cuboid is a right angled parllelo pipped). Its length 20 *m.* When 18 *l*. water is taken out of the water body, its level goes down by 15 *cm.* Find the breadth of the water body.

    (***Hint*** : 1 Litre = 1000 *ml*, 1 *ml* = 1 cm^3)
11. A 10 meter long, 4 meter high and 24 cm thick wall is to be constructed in an open ground. If the wall is to be built using bricks of the size 24 *cm* × 12 *cm* × 8 *cm*, then how many bricks will be needed?

12. Find out the expenditure on digging a cuboidal pit of 8 *m* length, 6 *m* width and 3 *m* depth. If the cost of digging is Rs. 30 per cubic meter.

13. The dimensions of a godown are 60 *m* × 25 *m* × 10 *m* What is the maximum number of wooden crates of dimensions 1.5 *m* × 1.25 *m* × 0.5 *m* that can be stacked in this godown?

14. A solid cube of side 12 *cm* is cut into 8 identical cubes each having the same volume. What will be the side of the new cubes? Also find out the surface area of both the cubes?

15. Mary wants to keep her christmas tree in a wooden box covered with coloured paper. She wants to know how much paper she needs to buy. The above mentioned box is 80 *cm* long, 40 *cm* wide and 20 *cm* high. And if the paper is 40 *cm* square sheets then how many sheets are needed.



What Have We Learnt



1. Cube is a regular hexagonal solid object. It has 4 lateral faces and 8 vertices.
2. If the side of the cube is of length '*a*' then

   The total surface area of the cube $= 6a^2$

   Lateral surface area of the cube $= 4a^2$

   Volume of the cube $= a^3$
3. If the length of a cuboid is '*l*', breadth '*b*' and height '*h*' then

   Total surface area of the cuboid $= 2(lb+bh+lh)$

   Lateral surface area of the cuboid $= 2h\ (l+ b)$

   Volume of the cuboid $= lbh$
4. Solid objects that have more volume occupy more space.